

Hypertension Guidelines

Understanding Hypertension

Hypertension, commonly known as high blood pressure, is a condition where the force of blood against artery walls is consistently too high. Often called the "silent killer," it typically has no symptoms but can lead to serious health complications if left untreated.

Blood Pressure Classifications

Normal Blood Pressure

- **Systolic:** Less than 120 mmHg
- **Diastolic:** Less than 80 mmHg
- **Reading:** <120/80 mmHg

Elevated Blood Pressure

- **Systolic:** 120-129 mmHg
- **Diastolic:** Less than 80 mmHg
- **Action:** Lifestyle modifications recommended

Stage 1 Hypertension

- **Systolic:** 130-139 mmHg OR
- **Diastolic:** 80-89 mmHg
- **Action:** Lifestyle changes and possibly medication

Stage 2 Hypertension

- **Systolic:** 140 mmHg or higher OR
- **Diastolic:** 90 mmHg or higher
- **Action:** Lifestyle changes and medication

Hypertensive Crisis

- **Systolic:** Higher than 180 mmHg AND/OR
- **Diastolic:** Higher than 120 mmHg
- **Action:** Seek emergency medical care immediately

Risk Factors

Non-Modifiable Risk Factors

- **Age:** Risk increases with age (especially over 65)
- **Family history:** Genetic predisposition
- **Race:** Higher prevalence in African Americans
- **Gender:** Men before age 55, women after menopause

Modifiable Risk Factors

- Obesity or being overweight
- Lack of physical activity
- High sodium diet
- Excessive alcohol consumption
- Tobacco use
- Chronic stress
- Poor sleep quality
- Certain chronic conditions (diabetes, kidney disease)

Symptoms and Complications

Warning Signs (Rare)

Most people with hypertension have no symptoms, but severe cases may cause:

- Severe headaches
- Nosebleeds
- Shortness of breath
- Chest pain
- Visual changes
- Blood in urine

Long-Term Complications

- **Heart disease:** Heart attack, heart failure
- **Stroke:** Reduced blood flow to brain
- **Kidney damage:** Chronic kidney disease
- **Vision loss:** Retinopathy
- **Peripheral artery disease:** Reduced blood flow to limbs
- **Cognitive decline:** Memory problems, dementia

Lifestyle Modifications (First-Line Treatment)

1. DASH Diet (Dietary Approaches to Stop Hypertension)

Foods to Emphasize:

- Fruits and vegetables (4-5 servings each daily)
- Whole grains (6-8 servings daily)
- Low-fat dairy (2-3 servings daily)
- Lean proteins (fish, poultry, beans)
- Nuts and seeds (4-5 servings weekly)

Foods to Limit:

- Sodium (less than 2,300 mg/day, ideally 1,500 mg/day)
- Saturated fats and cholesterol
- Red meat
- Sugary foods and beverages
- Processed foods

2. Sodium Reduction Strategies

Reading Labels:

- Check sodium content per serving
- Look for "low sodium" (140 mg or less per serving)
- Avoid "high sodium" foods (400 mg or more per serving)

Cooking Tips:

- Use herbs and spices instead of salt
- Rinse canned foods to remove excess sodium
- Cook from scratch when possible
- Limit condiments and sauces
- Choose fresh or frozen vegetables over canned

3. Physical Activity

Recommendations:

- **Aerobic exercise:** 150 minutes moderate or 75 minutes vigorous per week
- **Strength training:** 2 days per week
- **Examples:** Brisk walking, swimming, cycling, dancing

Getting Started:

- Start slowly and gradually increase intensity
- Find activities you enjoy
- Exercise with a friend for motivation
- Break activity into 10-minute sessions if needed
- Check with doctor before starting new program

4. Weight Management

Target Goals:

- Achieve and maintain healthy BMI (18.5-24.9)
- Lose 5-10% of body weight if overweight
- Reduce waist circumference (men <40 inches, women <35 inches)

Strategies:

- Track calorie intake
- Eat smaller portions
- Avoid emotional eating
- Stay hydrated
- Get adequate sleep

5. Alcohol Moderation

Recommended Limits:

- **Men:** No more than 2 drinks per day
- **Women:** No more than 1 drink per day

One Standard Drink:

- 12 oz beer (5% alcohol)
- 5 oz wine (12% alcohol)
- 1.5 oz spirits (40% alcohol)

6. Smoking Cessation

Benefits:

- Blood pressure drops within 20 minutes of quitting
- Heart attack risk decreases within 24 hours
- Long-term cardiovascular risk significantly reduced

Resources:

- Nicotine replacement therapy

- Prescription medications
- Counseling and support groups
- Quitlines: 1-800-QUIT-NOW

7. Stress Management

Techniques:

- Deep breathing exercises
- Meditation and mindfulness
- Yoga or tai chi
- Regular exercise
- Adequate sleep (7-9 hours)
- Time management
- Social support

Home Blood Pressure Monitoring

Choosing a Monitor

- Use validated, automatic upper-arm cuff monitors
- Ensure proper cuff size
- Avoid wrist or finger monitors

Proper Measurement Technique

1. **Prepare:** Don't smoke, exercise, or consume caffeine 30 minutes before
2. **Rest:** Sit quietly for 5 minutes before measuring
3. **Position:** Sit with back supported, feet flat, arm at heart level
4. **Measure:** Take 2-3 readings 1 minute apart
5. **Record:** Log date, time, and readings
6. **Timing:** Measure at same time daily (morning and evening)

When to Measure

- Before taking medications
- At the same times each day
- When experiencing symptoms
- As directed by healthcare provider

Medication Management

Common Blood Pressure Medications

1. Diuretics (Water Pills)

- Remove excess sodium and water
- Examples: Hydrochlorothiazide, Chlorthalidone
- Side effects: Increased urination, low potassium

2. ACE Inhibitors

- Relax blood vessels
- Examples: Lisinopril, Enalapril
- Side effects: Dry cough, elevated potassium

3. ARBs (Angiotensin II Receptor Blockers)

- Block effects of angiotensin
- Examples: Losartan, Valsartan
- Side effects: Dizziness, elevated potassium

4. Calcium Channel Blockers

- Relax blood vessel muscles
- Examples: Amlodipine, Diltiazem
- Side effects: Swelling, constipation

5. Beta Blockers

- Reduce heart rate and workload
- Examples: Metoprolol, Atenolol
- Side effects: Fatigue, cold hands/feet

Medication Adherence Tips

- Take medications at the same time daily
- Use pill organizers
- Set phone reminders
- Link to daily routine (e.g., brushing teeth)
- Never stop medications without consulting doctor
- Report side effects promptly

Special Populations

Pregnancy (Gestational Hypertension)

- Regular prenatal monitoring

- May require medication adjustment
- Increased risk of preeclampsia
- Close postpartum follow-up

Elderly Patients

- Target BP may be slightly higher (150/90 mmHg)
- Increased fall risk with aggressive treatment
- More medication interactions
- Regular medication review

Diabetes and Hypertension

- Target BP: <130/80 mmHg
- Increased cardiovascular risk
- ACE inhibitors or ARBs preferred
- Comprehensive risk factor management

Regular Health Monitoring

Recommended Screenings

- **Blood pressure:** Every visit or at least annually
- **Cholesterol:** Every 4-6 years (more often if abnormal)
- **Blood glucose:** Every 3 years (more often if at risk)
- **Kidney function:** Annually if on medications
- **Eye exam:** Annually if diabetic or complications
- **EKG:** As recommended by doctor

Emergency Situations

Seek Immediate Medical Attention If:

- Blood pressure >180/120 mmHg with symptoms
- Severe headache
- Chest pain or pressure
- Difficulty breathing
- Severe anxiety
- Nosebleed that won't stop
- Severe dizziness
- Vision changes
- Weakness or numbness

Action Plan Template

Daily Checklist

- ☐ Take medications as prescribed
- ☐ Monitor blood pressure (if recommended)
- ☐ Eat DASH diet meals
- ☐ Limit sodium intake
- ☐ Exercise for 30 minutes
- ☐ Practice stress reduction
- ☐ Avoid tobacco and limit alcohol

Weekly Goals

- ☐ Track weight
- ☐ Review blood pressure log
- ☐ Plan healthy meals
- ☐ Schedule physical activities
- ☐ Assess stress levels

Monthly Review

- ☐ Average blood pressure readings
- ☐ Weight trends
- ☐ Medication refills needed
- ☐ Schedule doctor appointments
- ☐ Adjust goals as needed

Important Reminders

⚠ **Medical Disclaimer:** This guide is for educational purposes only. Always consult your healthcare provider for personalized medical advice and treatment.

☐ **Emergency:** Call 911 immediately if experiencing chest pain, severe headache with confusion, or other emergency symptoms.

☐ **Medications:** Never stop or change blood pressure medications without consulting your doctor, even if you feel fine.

☐ **Regular Checkups:** Maintain regular appointments with your healthcare provider for blood pressure monitoring and medication adjustments.